

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-12 07:00 UTC

SEC P vs NP — notebook summary

Generated 2026-05-12 07:00 UTC

- Cycles recorded: **645**
- Time span: 441.3h (~1.46 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	608
BARRIER_HIT	18
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	34
Schur-Weyl Duality	22
Representation Theory of Symmetric Groups	17
Matroid Theory	15
Free Probability	14
Additive Combinatorics	10
Free Probability Theory	10
Noncommutative L^p Geometry	9
Polymatroid Theory	8
Algebraic Geometry	7
Random Matrix Theory	7
Finite Geometry	6
Spectral Graph Theory	6
Plethysm Theory	6
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4

Field	Cycles
FOURIER_ANALYSIS	4
Noncommutative Geometry	4
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3
Schur-Weyl duality	3
Diophantine Approximation	3
Persistent Homology	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-12 01:18 UTC	INCONCLUSIVE	Free Cumulant Sum Bounded by Disjointness
2026-05-12 01:34 UTC	INCONCLUSIVE	Communication Complexit Plethysm Coefficient Gap in Permanent vs Determinant Decompositio
2026-05-12 01:48 UTC	INCONCLUSIVE	Symmetric Group Fourier Coefficients Inverse Proportional to ACC ⁰
2026-05-12 01:54 UTC	INCONCLUSIVE	Integer Point Count in Solution Polytope Bounds Extended Frege Pr
2026-05-12 02:14 UTC	INCONCLUSIVE	Matroid Rank Bounded by Nisan-Wigderson Seed Length for 3-SAT
2026-05-12 02:20 UTC	INCONCLUSIVE	Matroid Rank Submodularity and Monotone Circuit Size for k-CLIQUE
2026-05-12 02:46 UTC	INCONCLUSIVE	Symplectic Rank of Disjointness Communication Matrix Bounds Rando
2026-05-12 03:10 UTC	INCONCLUSIVE	Schur Coefficient Density Inverse Proportional to Frege Proof Len
2026-05-12 03:25 UTC	INCONCLUSIVE	Free Cumulant Sum Bounded by Log-Size for Read-Twice BPs
2026-05-12 04:34 UTC	INCONCLUSIVE	Condition Number of Moment Matrix Inversely Proportional to SOS D
2026-05-12 04:40 UTC	INCONCLUSIVE	Projective Plane Incidence Bounds Nisan-Wigderson Seed Length

Time	Verdict	Title
2026-05-12 05:17 UTC	INCONCLUSIVE	Fourier Coefficient Spread and ACC ⁰ Circuit Lower Bounds
2026-05-12 05:43 UTC	INCONCLUSIVE	Fourier Coefficient Decay Bounds AC ⁰ Circuit Size for Parity-Inse
2026-05-12 06:36 UTC	INCONCLUSIVE	Invariant Ring Degree Bounds SOS Refutation Degree for Symmetric
2026-05-12 06:43 UTC	INCONCLUSIVE	Noncommutative L ^p Norm of Communication Matrix Bounds Randomized

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```